

Authenticated encryption chip implementation against side-channel attack

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Abstract. An authenticated encryption chip with novel nonce generation circuit was developed. This circuit associates the nonce generation with the contents and receiving time of plaintext. It will also generate overlapping power compensation for the whole chip. The integrated chip was fabricated under SMIC 180nm technology. Under 1.8V V_{DD} , 100MHz global clock, the power consumption is about 14mW within 50k gates. The test result exhibits more than 10 times the strength in resistance to side-channel attack than the unprotected version without increasing hardware cost.

1. Introduction

Side-Channel Attacks (SCA) is one of the most important security problems faced by authenticated encryption (AE) hardware implementation. When the cryptographic implementation works it is unavoidable that information leaks. An attacker can then exploit the leakage and destroy the confidentiality of data. Side-channels that can leak information for instance time, electromagnetic, and power. Among them, power analysis and electromagnetic analysis are most commonly used. Their principles are the same, so it is enough to analyse one of them. To reveal the secret key of hardware implementation, they both analyse a large number of traces that have been recorded while the device is processing[1]. They are universally applicable methods since they have the advantages that no in-depth knowledge of the device is required. When the implementation is functioning, the internal information processing will have a certain connection with the power consumption. The attacker can deduce the hardware internal data information by analysing the power or electromagnetic changes of the implementation, then crack the secret key[2]. The two most common way of power analysis are Differential Power Analysis (DPA) and Correlation Power Analysis (CPA). The former analyses the rationality of guessing the key by calculating the differential value. The latter cracks the key by calculating the correlation coefficient.

In this article, we chose the unprotected version of ASCON as authenticated encryption prototype. An authenticated encryption chip with power compensation and novel nonce generation circuit was generated. Two methods, “Reversed Grouping Padding (RGP)” and “Appended Random Delay (ARD)” were proposed to improve hardware resistance to SCA. These methods were realized by modifying the nonce generation mechanism without adding complex peripheral circuits. Our integrated chip was

fabricated under SMIC 180nm technology. The test result exhibits 10 times the strength in resistance to side-channel attack than the unprotected version with few increase in hardware cost.

2. Side-channel attack on ASCON

2.1. Basic theory of ASCON

The mode of operation of ASCON is based on duplex sponge modes like MonkeyDuplex. It is a well-known authentication encryption algorithm with superior performance for lightweight cases. ASCON contains four phases: *Initialization*, *Associated Data*, *Encryption/Decryption*, and *Finalization*, as figure 1 shows. The round transformation is a sponge-like SPN-based permutation p . Each round p contains constant-addition operation p_c , substitution layer p_s and linear diffusion layer p_L . Since the implementation of ASCON is not the focus of this article, here is only a brief introduction. Those who are interested in related content can refer to [3].

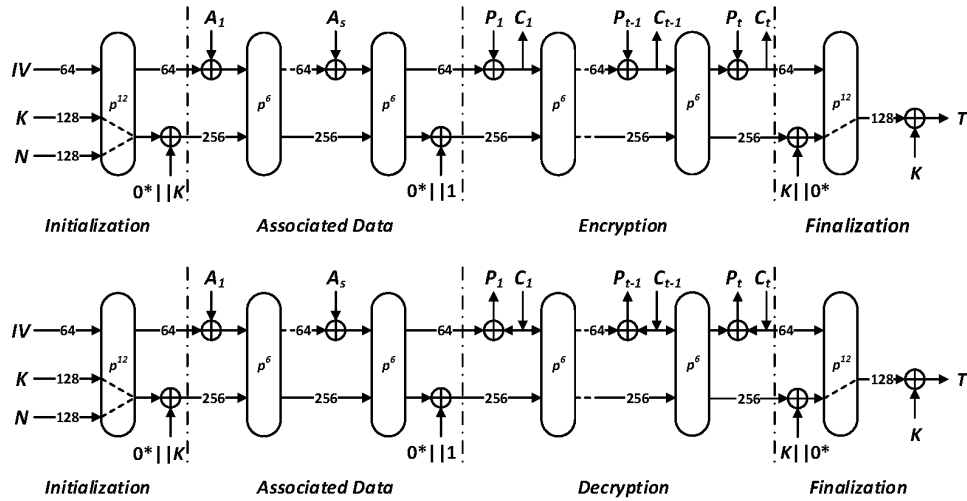


Figure 1. mode of operation of ASCON

2.2. Related work

An unprotected hardware implementation of ASCON and a Threshold Implementation (TI) variants are given in [4]. They were completed to the back-end routing stage under 90nm UMC technology with global clock of 1MHz and 1V power supply. For the unprotected version, about 1000 traces are needed for a successful attack. With about 3100 traces the success rate is higher than 99% on average. For the TI variants, SCA cannot seem to analyse the correct key. This masking method performs pretty well. However, the TI variants needs four times of power and area than the unprotected version.

The work of [1] is very systematic. It not only introduces the implementation of ASCON unprotected version, but also gives several SCA-protected variants, including Domain-Oriented Masking (DOM), Unified Masking (UMA) and Low-Latency Masking (LOLA). Designers can make a trade-off among area, throughput, randomness and latency according to the requirements of SCA protection and application scenarios. Since this work has a lot of crosswise tasks, each hardware implementation stays in front-end synthesis stage under 90nm UMC technology with global clock of 1MHz and 1V power supply. Although only netlist-level implementation results are available, this article still provides constructive guidance to designers who want to study the SCA protection of ASCON in detail.

[5] describes the SCA principle of ASCON. For the single S-box implementation, the non-linear transformation output is selected as the analysis point for the attack. For the multiple parallel S-box structure, the output of round transformation is taken as attack point. The masking protected version of ASCON needs about 150k traces to attack. Same as [4], about four times of power and hardware cost are needed for implementation.

Summarizing other state-of-the-art works, there have been many studies on the hardware implementation of ASCON, and there are also sufficient theories and schemes for its SCA research. These studies provide constructive support for our research. Our purpose is to proposed a chip-level implementation with certain safety improvements, but don't increase hardware cost.

2.3. Side-channel attack on ASCON

In order to improve hardware security, we need to analyse the SCA security risks of the algorithm. The most vulnerable phase is the *Initialization* for ASCON since three out of five input words (x_0, x_3, x_4) of state S are publicly known and the other two words (x_1, x_2) belong to the key. Therefore, the acknowledged power analysis attack on ASCON occurs in the *Initialization* phase. For 64 parallel S-boxes, it is impractical and inaccurate to analyse the output of a single nonlinear S-box separately. So, we chose the first whole round transformation of permutation p to analysis in a single clock cycle. Since the contents of the state at *Initialization* are known except for the key K (x_1, x_2), the attacker can vary the nonce N as different input and observe the output of the first-round transformation. Here we pick register x_0 as the analysis object to explain the attack on ASCON in detail.

The linear diffusion layer step for x_0 in ASCON is:

$$\sum_0 (x'_0) = x'_0 \oplus (x'_0 \ggg 19) \oplus (x'_0 \ggg 28) \quad (1)$$

Where x'_0 is the input of linear step p_L . The output of the non-linear S-box for x_0 is:

$$x'_0 = x_1x_4 + x_3 + x_1x_2 + x_2 + x_1x_0 + x_1 + x_0 \quad (2)$$

(2) can be rewritten to one quadratic term:

$$x'_0 = x_1(x_4 + x_2 + x_0 + 1) + x_3 + x_2 + x_0 \quad (3)$$

We know that x_0 is constant of the IV , x_1 and x_2 are constant of the key K , x_3 and x_4 are variable of the nonce N . We remove the words that add a constant amount to the activity to the register and the equation turns to:

$$x'_0 = x_1(x_4 + 1) + x_3 \quad (4)$$

Since each word x_i is 64 bits in ASCON, we combine equations (1) and (4) and change it into a register-like expression:

$$S_i(N, K^*) = k_0^*(n_{i+64} + 1) + n_i + k_1^*(n_{i+109} + 1) + n_{i+45} + k_2^*(n_{i+100} + 1) + n_{i+36} \quad (5)$$

From (5) we can see that one function contains 3 key bits thus there are 8 key guesses. In [6], a result shows that using fifty thousand traces results in a success rate of almost 1. That means we can obtain the key information of x_1 by attack on register x_0 , which is half of the secret key.

Similar to this derivation, we further analyse register x_1 . The linear diffusion layer step for x_1 in ASCON is:

$$\sum_1 (x'_1) = x'_1 \oplus (x'_1 \ggg 61) \oplus (x'_1 \ggg 39) \quad (6)$$

The output of the non-linear S-box for x_1 is:

$$x'_1 = x_2x_3 + x_4 + x_1x_3 + x_3 + x_1x_2 + x_2 + x_1 + x_0 \quad (7)$$

We change (7) like x_0 :

$$x'_1 = x_3(x_2 + x_1 + 1) + x_4 + x_2x_1 + x_2 + x_1 + x_0 \quad (8)$$

$$x'_1 = x_3(x_2 + x_1 + 1) + x_4 = x_3(x_{12} + 1) + x_4 \quad (9)$$

Combine (6) and (9), we get:

$$S_i(N, K^*) = n_i(k_0^* + 1) + n_{i+64} + n_{i+3}(k_1^* + 1) + n_{i+67} + n_{i+25}(k_2^* + 1) + n_{i+89} \quad (10)$$

Same as x_0 , we can attack on register x_1 to get the key candidates k^* . The difference is this time k^* does not represent the key itself, but the XOR of the corresponding key bits in the register x_1 and x_2 . The attacker cannot directly obtain the other half of the key x_2 , but can calculate x_2 through XOR of the known x_1 and x_{12} mediately. [6] shows that also fifty thousand traces are needed to get key guesses at a success rate of almost 1. That means by using power analysis, an attacker can possibly get all the key information from ASCON.

3. Detail of realization

The block diagram and data path of our integrated chip is shown in figure 2.

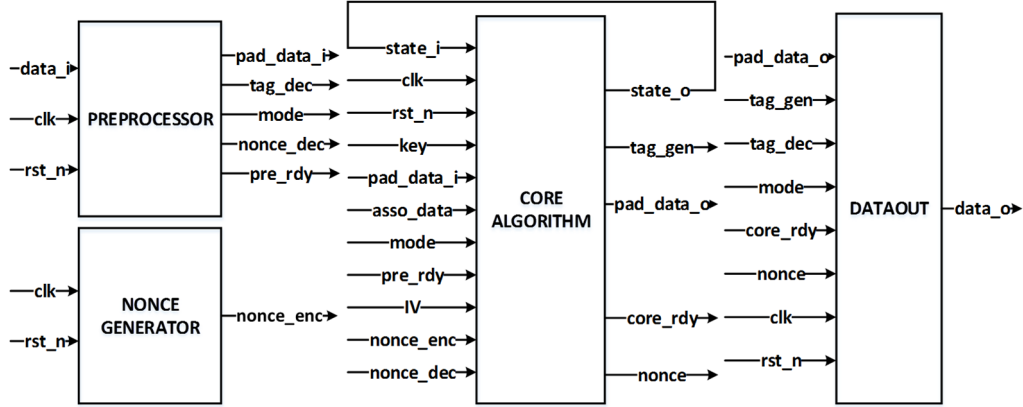


Figure 2. Block diagram of integrated chip

As analysed in section 2. The power analysis relies on the fact that the power dissipation of hardware is correlated to its activity, and associates with the key-dependent data being processed. Two mainstream countermeasures have been proposed in prior literature: hiding and masking[7]. After carefully studying the principles of these two methods, we noticed that the former is to hide the processed data into indistinguishable power patterns, which decreases the dependency between the power model and the physical dissipation[8]. The latter involves masking the key-dependent data with random variables, which attempts to reduce the dependency between the processed data and the power model mainly at algorithmic or gate level. The masking method usually needs several times hardware resources than the unprotected implementation[9]. In our design, hiding method was used as chip-level countermeasure since it saves hardware cost. It can also be used together with masking for further improvements.

We know that in practice, the *NONCE GENERATOR* module is needed to generate nonce N . This module contains a great deal of calculation processing, thus can provide the power patterns naturally. Therefore, we need *NONCE GENERATOR* to generate nonce and at the same time as a power compensation module. The real problem is that if the 128-bit nonce is directly generated, this part of the power compensation will have a simple repetitive rule, which can be easily eliminated by the attacker in a statistical way. Moreover, according to section 2, nonce needs to participate in ASCON's SCA as a known quantity. By analysing the law between power consumption and time, it is even possible for an attacker to infer the law of nonce generation, thereby inferring the value of nonce in each encryption process.

3.1. "Reversed Group Padding" method

In order to solve the existing problems, we came up with a "Reversed Group Padding (RGP)" method. The nonce generation is based on the principle of Linear Feedback Shift Register (LFSR). We didn't directly use 128-bit LFSR to form 128-bit Deterministic Random Bit Generators (DRBG) since the hardware cost is unfavourable. The method we take is shown in figure 3. Two 16-bit *DRBG* modules employs different seed and XOR logic. Once the chip is powered on, they start to work synchronously. Under the control of the *Counter*, the variational output of them are sent to 128-bit nonce simultaneously and reversely. That is why this method was named like this. In this case, hardware cost is reduced by about 70%. Two 16-bit generator's power is superimposed on each other, and the correlation with the nonce decreases. It is no longer possible for the attacker to infer the nonce generation rules by analysing the relationship between power consumption and time. Besides, this power compensation is of the same magnitude as the valid data, which is much larger than the noise magnitude. It is extremely difficult to eliminate the power compensation by statistical methods.

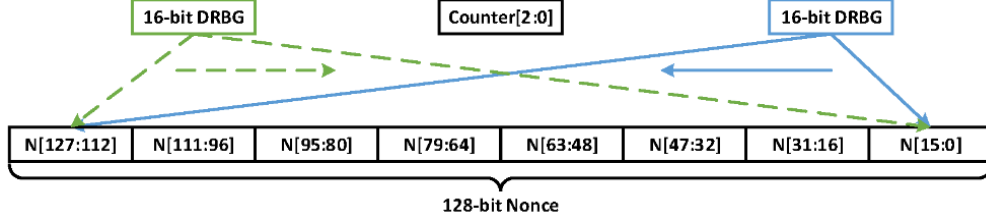


Figure 3. “Reversed Grouping Padding” method

3.2. “Appended Random Delay” sampling solution

On this basis, we further proposed an “Appended Random Delay (ARD)” sampling solution. We note that only one nonce value will be used in each encryption process. If the pseudo-random nonce is sampled at a fixed point each time and used for encryption, the attacker can easily find the rule. To deal with this problem, we proposed a nonce generation principle used in encryption as depicted in figure 4. Where, i, j, k are any three bits in the plaintext, which can be arbitrarily specified by the designer without affecting the throughput. During encryption, when the k^{th} bit of plaintext is detected, the start point of nonce sampling, T_S , is set, but sampling will not start at this time. Meanwhile, i^{th}, j^{th}, k^{th} bits of plaintext form an initial value for the counter and decrease to 0, resulting in a nonce sampling delay time T_D . Therefore, the real nonce is extracted at $T = T_S + T_D$. This series of calculation will not affect the throughput of the chip because it is performed in parallel during the data reception process. It can be seen that the nonce used in the encryption process not only relies on the arrival time of the plaintext, but the value of plaintext as well. For the attacker, the nonce becomes uncontrollable and this greatly increases the attack complexity.

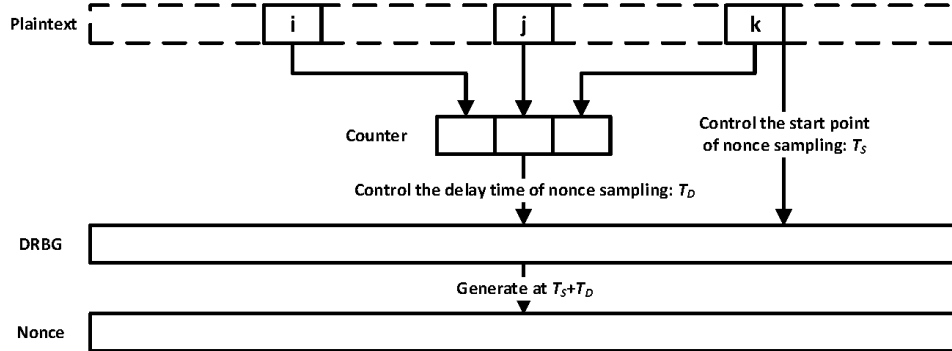


Figure 4. “Appended Random Delay” sampling solution

4. Results and discussion

We apply these designs to the unprotected ASCON AE hardware implementation. The whole AE chip with certain protection was fabricated under SMIC 180nm technology and packaged in SOP-28. The optical photo of chip is shown in figure 5. Under 1.8V V_{DD} and 100MHz global clock, the chip exhibits a power consumption of about 14mW within 50k gates. In this case, it reaches a high throughput of approximately 1Gbps, which is the same as the unprotected version. This proved that these methods will not affect throughput of the implementation. Besides, side-channel attack was presented on the implemented chip. As mentioned in section 3, we have made RGP and ARD treatment for power compensation and nonce generation. The attacker cannot directly obtain the nonce corresponding to each traces of power analysis. In order to proceed the attack, we lowered the difficulty by letting the attacker know the nonce corresponding to each round of trace generation. The test method was stated in subsection 2.3. We applied 64 attacks on x_0 and 64 attacks on x_1 to analyse all 128 bits of possible key. Each attack corresponds to 8 possible keys, which means 8 different correlation traces could be obtained. As an example, figure 6 shows the correlation traces corresponding to the first 9 bits analysis of x_0 . The trace with highest correlation is marked in red, which correspond to the most likely correct key, and the

others are marked in grey. It's easy to recognize that the trace with highest correlation have no significant differences with other traces. In addition, the maximum value in the red curve is not prominent, which indicates that the sampling point has little correlation with the statistical calculated value, thus cannot reliably crack the correct key.

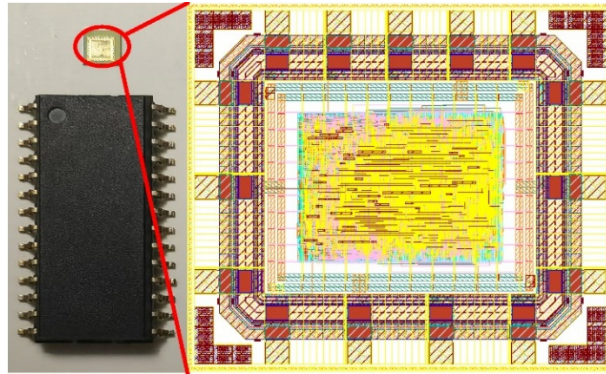


Figure 5. Optical photo of chip

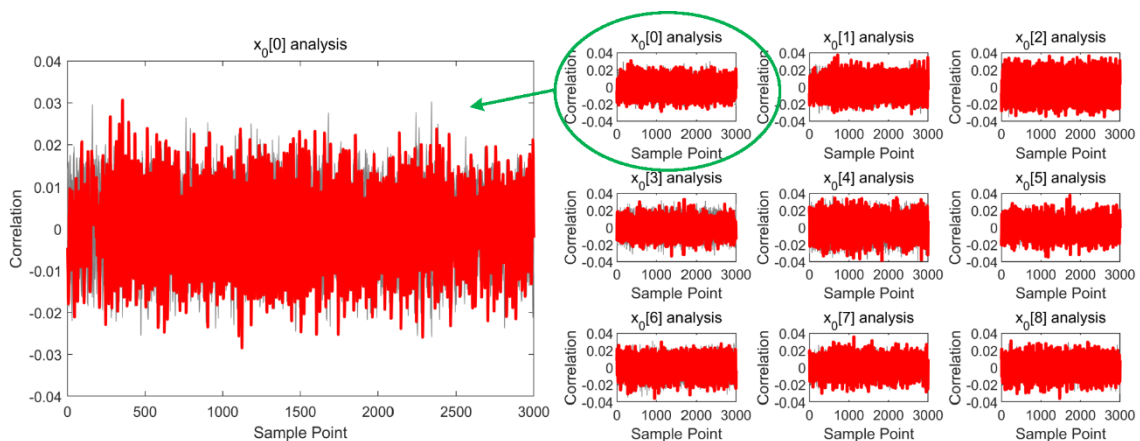


Figure 6. Correlation-based electromagnetic analysis

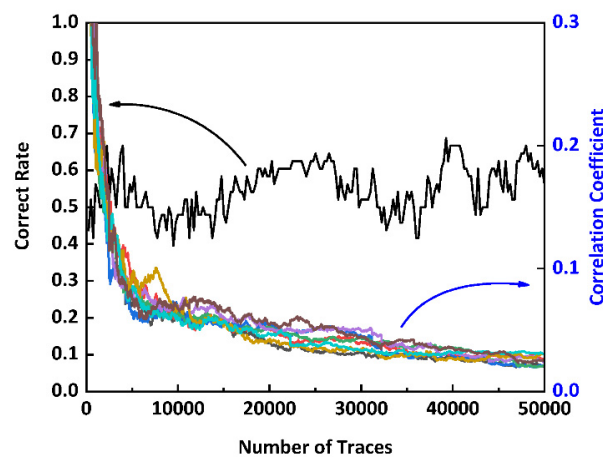


Figure 7. Correct rate and correlation coefficient vs. number of traces

We further calculated the correlation coefficient and guessing correct rate for all guessed key under different number of traces. The results are depicted in figure 7. The curve above the image shows the relationship between statistical speculation accuracy of all guess keys and trace number, which corresponds to the left Y axis. The curves group below the image reveals the relationship between correlation coefficient and trace number. Each curve corresponds to a key guess and the ordinate matches the right Y axis. It can be seen that the statistical accuracy of key guessing is maintained at about 50%, and there is no obvious difference between the correlation coefficients under different guessing keys. These results show that as high as 50k traces is not enough to cause the leakage of key information for our proposed chip. While in [4], however, the unprotected implementation only needs about 3100 traces to achieve crack success rate greater than 99%. By the way, these methods will not increase additional hardware consumption since the number of registers for nonce generation has been reduced from 128 to 38.

5. Conclusions

An authenticated encryption chip with novel nonce generation circuit was developed. This integrated chip was fabricated under SMIC 180nm technology. The power consumption is about 14mW within 50k gates under 1.8V V_{DD} , 100MHz global clock. Two methods, RGP and ARD was proposed and applied for power compensation and nonce generation. The test result exhibits 10 times the strength in resistance to side-channel attack than the unprotected version. This method is easy to implement. Although it cannot completely solve the SCA security problem in principle, it can create lots of difficulties for the attacker's analysis process. Most importantly, it does not cause additional hardware cost and can be shared with other SCA countermeasures. It has a wide range of application value in encryption and security hardware implementations.

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